

Лабораторная работа №16

Базовая защита от атак типа «brute force»

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Информация

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Вводная часть

- Целью данной работы является приобретение навыков настройки сервера NFS для удалённого доступа к ресурсам.

Основная часть

Защита с помощью Fail2ban

- Установка на сервере fail2ban.

```
[arromichina@server.arromichina.net ~]$ sudo -i
[sudo] password for arromichina:
[root@server.arromichina.net ~]# dnf -y install fail2ban
Extra Packages for Enterprise Linux 10 - x86_64
Extra Packages for Enterprise Linux 10 - x86_64
Rocky Linux 10 - BaseOS
Rocky Linux 10 - BaseOS
Rocky Linux 10 - AppStream
Rocky Linux 10 - AppStream
Rocky Linux 10 - CRB
Rocky Linux 10 - CRB
Rocky Linux 10 - Extras
Rocky Linux 10 - Extras
Dependencies resolved.
```

```
56 kB/s | 36 kB      00:00
4.7 MB/s | 5.5 MB   00:01
12 kB/s | 4.3 kB    00:00
5.0 MB/s | 5.3 MB   00:01
14 kB/s | 4.3 kB    00:00
2.8 MB/s | 2.0 MB   00:00
12 kB/s | 4.3 kB    00:00
613 kB/s | 486 kB   00:00
9.8 kB/s | 3.1 kB    00:00
12 kB/s | 4.8 kB    00:00
```

```
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
fail2ban                               noarch            1.1.0-6.el10_0    epel               9.4 k
Installing dependencies:
fail2ban-firewalld                     noarch            1.1.0-6.el10_0    epel               9.6 k
fail2ban-selinux                       noarch            1.1.0-6.el10_0    epel               31 k
fail2ban-sendmail                      noarch            1.1.0-6.el10_0    epel               12 k
fail2ban-server                        noarch            1.1.0-6.el10_0    epel              561 k
=====
```

- Запуск сервера fail2ban.

```
[root@server.arromichina.net ~]# systemctl start fail2ban
[root@server.arromichina.net ~]# systemctl enable fail2ban
Created symlink '/etc/systemd/system/multi-user.target.wants/fail2ban.service' → '/usr/lib/systemd/system/fail2ban.service'.
[root@server.arromichina.net ~]# █
```

- Запуск просмотра в дополнительном терминале журнала событий fail2ban.

```
[root@server.arromichina.net ~]# tail -f /var/log/fail2ban.log
2025-12-14 16:49:25,673 fail2ban.server [13755]: INFO -----
2025-12-14 16:49:25,673 fail2ban.server [13755]: INFO Starting Fail2ban v1.1.0
2025-12-14 16:49:25,675 fail2ban.observer [13755]: INFO Observer start...
2025-12-14 16:49:25,687 fail2ban.database [13755]: INFO Connected to fail2ban persistent database '/var/lib/fai
l2ban/fail2ban.sqlite3'
2025-12-14 16:49:25,690 fail2ban.database [13755]: WARNING New database created. Version '4'
```

- Создание файла с локальной конфигурацией fail2ban.

```
[root@server.arromichina.net ~]# touch /etc/fail2ban/jail.d/customisation.local  
[root@server.arromichina.net ~]#
```

Защита с помощью Fail2ban

- Настройка в файле `/etc/fail2ban/jail.d/customisation.local` времени блокирования на 1 час и включение защиты SSH.

```
customisation.local  [-M--] 14 L:[ 1+12 13/ 13] *(150 / 150b) <EOF>
[DEFAULT]
bantime = 3600
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

- Перезапуск fail2ban.

```
[root@server.arromichina.net ~]# systemctl restart fail2ban  
[root@server.arromichina.net ~]#
```

Защита с помощью Fail2ban

- Просмотр журнала событий.

```
2025-12-14 16:51:37,369 fail2ban.filter [14109]: INFO maxRetry: 5
2025-12-14 16:51:37,369 fail2ban.filter [14109]: INFO findtime: 600
2025-12-14 16:51:37,369 fail2ban.actions [14109]: INFO banTime: 3600
2025-12-14 16:51:37,370 fail2ban.filter [14109]: INFO encoding: UTF-8
2025-12-14 16:51:37,370 fail2ban.jail [14109]: INFO Creating new jail 'selinux-ssh'
2025-12-14 16:51:37,389 fail2ban.jail [14109]: INFO Jail 'selinux-ssh' uses pyinotify {}
2025-12-14 16:51:37,395 fail2ban.jail [14109]: INFO Initiated 'pyinotify' backend
2025-12-14 16:51:37,397 fail2ban.datedetector [14109]: INFO date pattern '': 'Epoch'
2025-12-14 16:51:37,397 fail2ban.filter [14109]: INFO maxRetry: 5
2025-12-14 16:51:37,398 fail2ban.filter [14109]: INFO findtime: 600
2025-12-14 16:51:37,398 fail2ban.actions [14109]: INFO banTime: 3600
2025-12-14 16:51:37,398 fail2ban.filter [14109]: INFO encoding: UTF-8
2025-12-14 16:51:37,408 fail2ban.filter [14109]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 0, has
h = 44e80c84ce5435ceab93026f64322873b6a6609)
2025-12-14 16:51:37,414 fail2ban.jail [14109]: INFO Creating new jail 'sshd-ddos'
2025-12-14 16:51:37,414 fail2ban.jail [14109]: INFO Jail 'sshd-ddos' uses pyinotify {}
2025-12-14 16:51:37,425 fail2ban.jail [14109]: INFO Initiated 'pyinotify' backend
2025-12-14 16:51:37,429 fail2ban.filter [14109]: INFO maxLines: 1
2025-12-14 16:51:37,432 fail2ban.filter [14109]: INFO maxRetry: 5
2025-12-14 16:51:37,432 fail2ban.filter [14109]: INFO findtime: 600
2025-12-14 16:51:37,434 fail2ban.actions [14109]: INFO banTime: 3600
2025-12-14 16:51:37,434 fail2ban.filter [14109]: INFO encoding: UTF-8
2025-12-14 16:51:37,438 fail2ban.jail [14109]: INFO Jail 'sshd' started
2025-12-14 16:51:37,444 fail2ban.jail [14109]: INFO Jail 'selinux-ssh' started
2025-12-14 16:51:37,453 fail2ban.jail [14109]: INFO Jail 'sshd-ddos' started
2025-12-14 16:51:37,464 fail2ban.filter [14109]: WARNING [selinux-ssh] Ignoring all log entries older than 600s;
these are probably messages generated while fail2ban was not running.
2025-12-14 16:51:37,464 fail2ban.filter [14109]: WARNING [selinux-ssh] Please check a jail for a timing issue. L
ine with odd timestamp: type=USER_END msg=audit(1765730497.441:536): pid=7148 uid=1000 auid=1000 ses=2 subj=unconfined_u
:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:session_close grantors=pam_keyinit,pam_limits,pam_systemd,pam_unix
```

Защита с помощью Fail2ban

- Включение защиты HTTP в файле `/etc/fail2ban/jail.d/customisation.local`.

```
#  
# HTTP servers  
#  
[apache-auth]  
enabled = true  
[apache-badbots]  
enabled = true  
[apache-noscript]  
enabled = true  
[apache-overflows]  
enabled = true  
[apache-nohome]  
enabled = true  
[apache-botsearch]  
enabled = true  
[apache-fakegooglebot]  
enabled = true  
[apache-modsecurity]  
enabled = true  
[apache-shellshock]
```

- Перезапуск fail2ban.

```
[root@server.arromichina.net ~]# systemctl restart fail2ban  
[root@server.arromichina.net ~]# █
```

- Просмотр журнала событий.

```
[root@server.arromichina.net ~]# tail -f /var/log/fail2ban.log
2025-12-14 16:53:13,056 fail2ban.jail      [14192]: INFO      Jail 'apache-auth' started
2025-12-14 16:53:13,058 fail2ban.jail      [14192]: INFO      Jail 'apache-badbots' started
2025-12-14 16:53:13,061 fail2ban.jail      [14192]: INFO      Jail 'apache-noscript' started
2025-12-14 16:53:13,063 fail2ban.jail      [14192]: INFO      Jail 'apache-overflows' started
2025-12-14 16:53:13,067 fail2ban.jail      [14192]: INFO      Jail 'apache-nohome' started
2025-12-14 16:53:13,069 fail2ban.jail      [14192]: INFO      Jail 'apache-botsearch' started
2025-12-14 16:53:13,072 fail2ban.jail      [14192]: INFO      Jail 'apache-fakegooglebot' started
2025-12-14 16:53:13,075 fail2ban.jail      [14192]: INFO      Jail 'apache-modsecurity' started
2025-12-14 16:53:13,078 fail2ban.jail      [14192]: INFO      Jail 'apache-shellshock' started
2025-12-14 16:53:13,081 fail2ban.jail      [14192]: INFO      Jail 'sshd-ddos' started
```

Защита с помощью Fail2ban

- Включение защиты почты в файле `/etc/fail2ban/jail.d/customisation.local`.

```
#  
# Mail servers  
#  
[postfix]  
enabled = true  
[postfix-rbl]  
enabled = true  
[dovecot]  
enabled = true  
[postfix-sasl]  
enabled = true
```

- Повторный перезапуск fail2ban.

```
[root@server.arromichina.net ~]# systemctl restart fail2ban  
[root@server.arromichina.net ~]# █
```

Защита с помощью Fail2ban

- Просмотр журнала событий.

```
[root@server.arromichina.net ~]# tail -f /var/log/fail2ban.log
2025-12-14 16:54:00,542 fail2ban.jail [14298]: INFO Jail 'apache-shellshock' started
2025-12-14 16:54:00,544 fail2ban.jail [14298]: INFO Jail 'postfix' started
2025-12-14 16:54:00,547 fail2ban.jail [14298]: INFO Jail 'postfix-rbl' started
2025-12-14 16:54:00,549 fail2ban.jail [14298]: INFO Jail 'dovecot' started
2025-12-14 16:54:00,551 fail2ban.jail [14298]: INFO Jail 'postfix-sasl' started
2025-12-14 16:54:00,555 fail2ban.jail [14298]: INFO Jail 'sshd-ddos' started
2025-12-14 16:54:00,571 fail2ban.filtersystemd [14298]: INFO [postfix-rbl] Jail is in operation now (process new journal entries)
2025-12-14 16:54:00,572 fail2ban.filtersystemd [14298]: INFO [postfix-sasl] Jail is in operation now (process new journal entries)
2025-12-14 16:54:00,573 fail2ban.filtersystemd [14298]: INFO [postfix] Jail is in operation now (process new journal entries)
2025-12-14 16:54:00,574 fail2ban.filtersystemd [14298]: INFO [dovecot] Jail is in operation now (process new journal entries)
```

Проверка работы Fail2ban

- Просмотр на сервере статуса fail2ban, статуса защиты SSH в fail2ban и установка максимального количества ошибок для SSH (=2).

```
[root@server.arromichina.net ~]# fail2ban-client status
Status
|- Number of jail:      16
`- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-modsecurity, apache-nohome,
apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix, postfix-rbl, postfix-sasl, selinux-ssh, sshd, ss
hd-ddos
[root@server.arromichina.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 0
| |- Total failed:    0
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:    0
   `-- Banned IP list:
[root@server.arromichina.net ~]# fail2ban-client set sshd maxretry 2
2
[root@server.arromichina.net ~]#
```

- Попытка зайти с клиента по SSH на сервер с неправильным паролем.

```
[root@client.arromichina.net ~]# ssh arromichina@server.arromichina.net
The authenticity of host 'server.arromichina.net (192.168.1.1)' can't be established.
ED25519 key fingerprint is SHA256:LUBVKU3n9LeYkjemiPx1VkJkq9Eq9laMvpgqUuTCjHA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added 'server.arromichina.net' (ED25519) to the list of known hosts.
arromichina@server.arromichina.net's password:
Permission denied, please try again.
arromichina@server.arromichina.net's password:
```

Проверка работы Fail2ban

- Просмотр на сервере статуса защиты SSH, разблокировка IP-адреса клиента и повторная проверка.

```
[root@server.arromichina.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 1
| |- Total failed:    3
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 1
   |- Total banned:    1
   `-- Banned IP list:  192.168.1.2
[root@server.arromichina.net ~]# fail2ban-client set sshd unbanip 192.168.1.2
1
[root@server.arromichina.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
| |- Currently failed: 1
| |- Total failed:    3
| `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
```

- Добавление в раздел по умолчанию игнорирование адреса клиента в конфигурационном файле `/etc/fail2ban/jail.d/customisation.local`.

```
customisation.local  [-M--] 34 L:[ 1+ 2 3/ 47] *(59 / 635b) 0010 0x00A
[DEFAULT]
bantime = 3600
ignoreip = 127.0.0.1/8 192.168.1.2
#
# SSH servers
#
```

- Перезапуск fail2ban.

```
[root@server.arromichina.net ~]# systemctl restart fail2ban  
[root@server.arromichina.net ~]# █
```

Проверка работы Fail2ban

- Просмотр журнала событий.

```
[root@server.arromichina.net ~]# tail -f /var/log/fail2ban.log
2025-12-14 17:06:06,336 fail2ban.jail [14701]: INFO Jail 'apache-shellshock' started
2025-12-14 17:06:06,337 fail2ban.filterssystemd [14701]: INFO [postfix] Jail is in operation now (process new journal
entries)
2025-12-14 17:06:06,339 fail2ban.jail [14701]: INFO Jail 'postfix' started
2025-12-14 17:06:06,341 fail2ban.filterssystemd [14701]: INFO [postfix-rbl] Jail is in operation now (process new journal
entries)
2025-12-14 17:06:06,342 fail2ban.jail [14701]: INFO Jail 'postfix-rbl' started
2025-12-14 17:06:06,343 fail2ban.filterssystemd [14701]: INFO [dovecot] Jail is in operation now (process new journal
entries)
2025-12-14 17:06:06,345 fail2ban.jail [14701]: INFO Jail 'dovecot' started
2025-12-14 17:06:06,346 fail2ban.filterssystemd [14701]: INFO [postfix-sasl] Jail is in operation now (process new journal
entries)
2025-12-14 17:06:06,347 fail2ban.jail [14701]: INFO Jail 'postfix-sasl' started
2025-12-14 17:06:06,349 fail2ban.jail [14701]: INFO Jail 'sshd-ddos' started
```

- Попытка войти с клиента на сервер с неправильным паролем.

```
[root@client.arromichina.net ~]# ssh arromichina@server.arromichina.net
arromichina@server.arromichina.net's password:
2Permission denied, please try again.
arromichina@server.arromichina.net's password:
```

- Просмотр статуса защиты SSH.

```
[root@server.arromichina.net ~]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:    0
|  `-- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:    0
   `-- Banned IP list:
```

Внесение изменений в настройки внутреннего окружения виртуальных машин

- Переход на виртуальной машине server в каталог для внесения изменений в настройки внутреннего окружения /vagrant/provision/server/, создание в нём каталога protect, в который помещаем в соответствующие подкаталоги конфигурационные файлы. Создание в каталоге /vagrant/provision/server исполняемого файла protect.sh.

```
[root@server.arromichina.net ~]# cd /vagrant/provision/server
[root@server.arromichina.net server]# mkdir -p /vagrant/provision/server/protect/etc/fail2ban/jail.d
[root@server.arromichina.net server]# cp -R /etc/fail2ban/jail.d/customisation.local /vagrant/provision/server/protect/etc/fail2ban/jail.d/
[root@server.arromichina.net server]# cd /vagrant/provision/server
[root@server.arromichina.net server]# touch protect.sh
[root@server.arromichina.net server]# chmod +x protect.sh
[root@server.arromichina.net server]#
```

Внесение изменений в настройки внутреннего окружения виртуальных машин

- Открытие файла на редактирование и добавление в него скрипта.

```
protect.sh      [-M--] 24 L:[ 1+ 9 10/ 10] *(280 / 280b) <EOF>
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Внесение изменений в настройки внутреннего окружения виртуальных машин

- Добавление конфигураций в конфигурационном файле Vagrantfile для сервера.

```
server.vm.provision :shell, :path => "provision/server/netlog.sh", :preserve_order => true, :type => "shell", :preserve_order => true, :path => "provision/server/protect.sh"
end
```

Вывод

- В ходе выполнения лабораторной работы были получены навыки работы с программным средством Fail2ban для обеспечения базовой защиты от атак типа «brute force».